

CBRE is a global leader in commercial real estate, managing thousands of buildings and providing consulting services in investment and management. They are based in over 100 countries and serve nearly 90 of the Fortune 100 companies. They have come to us to help them cut costs on their call centers that are set up to handle building maintenance and repairs. CBRE currently has a solution in place that automates their calls but their autoclassification is frequently wrong, for example it dials 911 for low severity issues. There is no human in the loop to intercept these issues and their exact failure rate is unknown.

This continued pattern will be costly financially with emergency contractor dispatches triggered by false high severity and misrouted tickets costing CBRE unnecessary trips. Meanwhile the paid call staff in the centers are not intervening before mistakes happen. Beyond the operational costs, the recurrent mishandling of client requests will erode their trust and values core to CBRE.

We will design a solution that will directly integrate with CBRE's current ticketing system which has a code or category in place for each type of maintenance issue. This means that we will not need to invent a new classification framework but instead will map to the current system. Additionally we know that we have call transcripts which we will need to parse and extract structured substance from with details such as building locations, problem types, indicators for severity, caller descriptions, and pair them with the routing decisions made and outcomes that followed. This will help give us a complete picture for each historical case knowing what the caller reported, where the system sent it, and the outcome of the routing. By indexing this past data and putting it into a searchable knowledge base, the system can retrieve past incidents that are similar and use their outcomes in classification. This foundation will build upon itself as more historical data becomes processed and improves the accuracy of the system.

Each new call made with this system can be processed through a series of steps involving extracting the keys from each transcript and retrieval from semantically similar historical calls. Next, classification and urgency can be determined from the caller's description and comparable past incidents. Each call will result in a routing recommendation. For routings with low confidence or unusual situations, a hybrid system can be firstly put into place with human intervention when necessary. This human in the loop will improve future outcomes by strengthening historical data and classifications. This naturally means that over time CBRE will be able to cut spending on these call centers and move toward more automated systems.

By combining intelligent classification with human oversight and a self improving data foundation, this solution will address both the cost problem and the underlying trust issue in the long run.